



IILM UNIVERSITY

Greater Noida, U.P.

INTERNSHIP PRESENTATION

Java Programming Internship

- STUDENT NAME **AMAN KUMAR**
- ENROLLMENT NUMBER **25SCS1003004673**
- PROGRAM & SEMESTER **B.Tech CSE – 3rd Semester**
- ORGANIZATION **CodeAlpha**
- INTERNSHIP DURATION **20 July 2026 – 20 August 2026**

School of Computer Science and Engineering

Internship Overview

A practical, assignment-oriented Java programming experience

Domain

Java Programming

Duration

20 July 2026 – 20 August 2026

Mode

Virtual / Assignment-Based

- The internship focused on applying Java programming concepts to practical programming activities.
- The experience emphasized problem solving, implementation, testing, debugging and structured submission.
- The internship helped connect academic programming concepts with applied software-development practices.

Organization Profile

CodeAlpha — internship and practical learning platform

- CodeAlpha provides virtual internship programs, practical coding assignments and project-oriented learning opportunities.
- The internship model is assignment-based, encouraging learners to gain practical exposure through hands-on work.
- Internship tracks cover multiple technical domains, including Java Programming.
- The journey generally involves receiving assignments, completing and submitting work, evaluation and certification after successful completion.
- The virtual structure also supports self-learning, time management and independent problem solving.

Problem Statement

Bridging the gap between programming theory and practical implementation

The Challenge

Students may understand syntax and definitions but still face difficulty when designing a solution, breaking a problem into steps, implementing logic and debugging errors.

Internship Approach

Practical, assignment-based programming work was used to apply Java concepts and improve the ability to develop, test and refine software solutions independently.

Objectives

Key goals of the internship

01

Java Fundamentals

02

OOP & Design

03

Code Quality

04

Debugging

05

Requirement Handling

06

Independent Work

Scope of the Internship

Technical and professional areas covered

Technical Scope

- Java programming fundamentals and syntax
- Object-oriented programming and structured program design
- Logical decomposition and step-by-step implementation
- Input/output handling and basic application concepts
- Testing, debugging and correcting programming errors
- Organized code and a clear development workflow

Professional Scope

- Working independently on assigned tasks
- Managing time during a fixed internship period
- Submitting completed work in a structured manner
- Improving communication and documentation skills
- Building practical confidence for future software-development projects

Technologies & Tools

Core technical focus

Primary Technology

JAVA
Programming language used
throughout the internship focus.

Core Concepts

OOP • Control Structures
Input / Output • Logic
Testing • Debugging

Tooling

Tools could vary by assignment. The
report identifies Java programming
and practical problem solving as the
consistent technical focus.

System Architecture / Workflow

A repeatable process for programming assignments



Key Principle

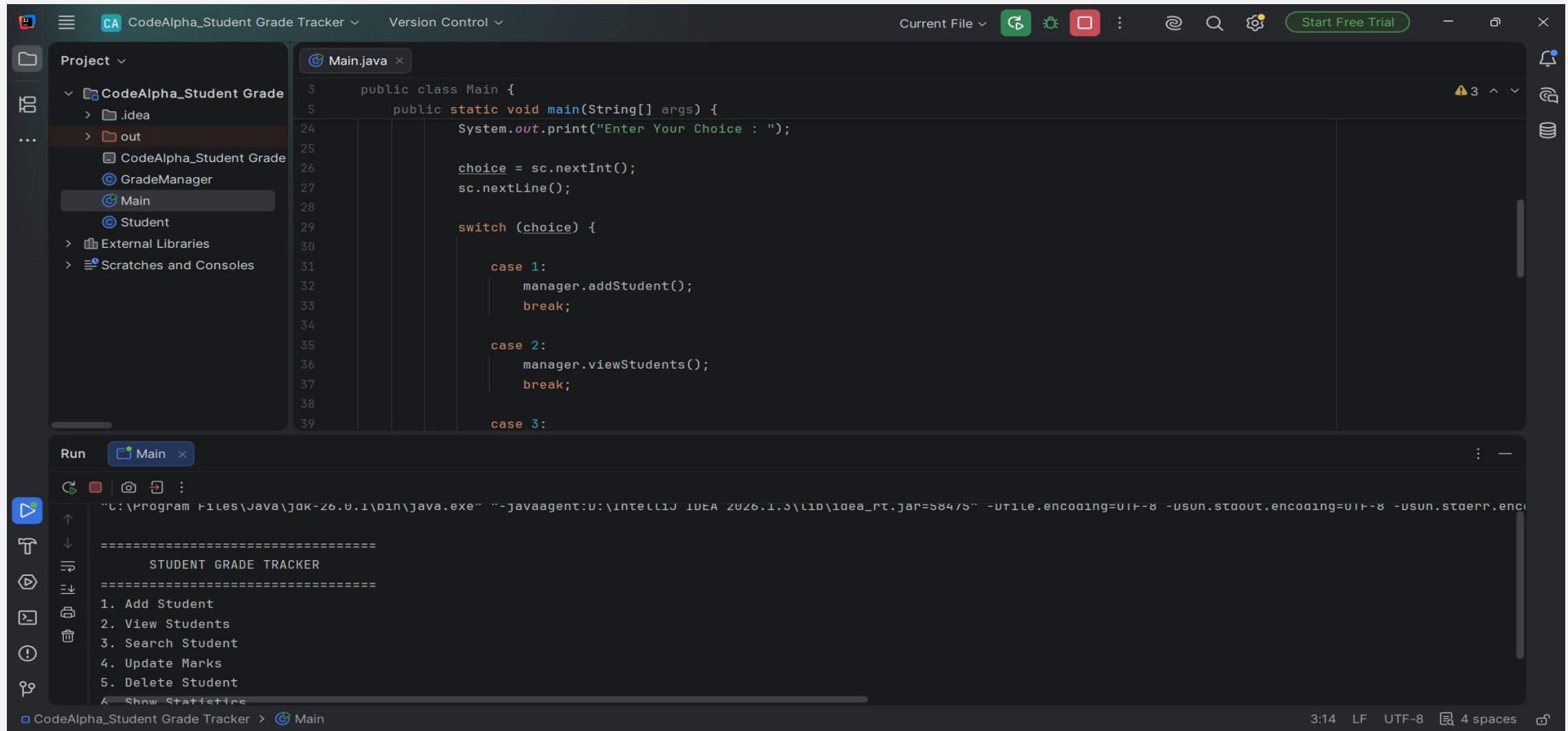
Writing the first version of code is not the finish line — testing, debugging and review are part of completing the task.

Methodology

Task-oriented approach followed during the internship

- | | | |
|-----------|-----------------------------------|--|
| 01 | Understand the Requirement | Identify inputs, expected output/behaviour and constraints. |
| 02 | Plan the Solution | Break the problem into manageable parts and choose suitable Java constructs. |
| 03 | Implementation | Convert the plan into readable and logically structured Java code. |
| 04 | Testing & Debugging | Run test inputs, identify syntax/runtime/logic issues and correct them. |
| 05 | Review & Documentation | Check organization, clarity and readiness before submission. |

Project Implementation (Student Grade Tracker)



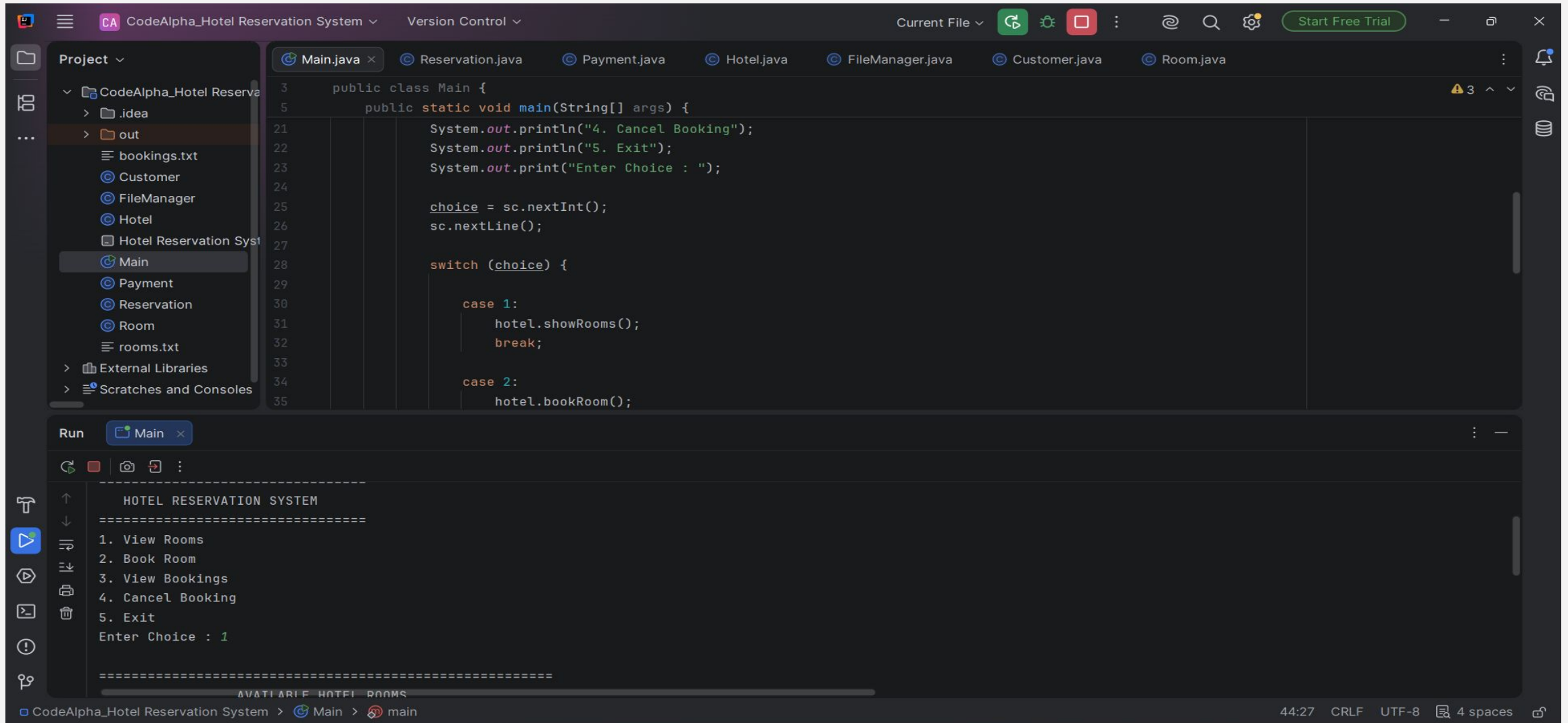
The screenshot shows an IDE with the following components:

- Project View:** Shows the project structure for "CodeAlpha_Student Grade Tracker". The "Main" class is selected.
- Main.java:** Contains the following code:

```
3 public class Main {  
5     public static void main(String[] args) {  
24         System.out.print("Enter Your Choice : ");  
25  
26         choice = sc.nextInt();  
27         sc.nextLine();  
28  
29         switch (choice) {  
30  
31             case 1:  
32                 manager.addStudent();  
33                 break;  
34  
35             case 2:  
36                 manager.viewStudents();  
37                 break;  
38  
39             case 3:
```
- Run Window:** Shows the output of the program:

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=58475" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8  
=====   
STUDENT GRADE TRACKER  
=====   
1. Add Student  
2. View Students  
3. Search Student  
4. Update Marks  
5. Delete Student  
6. Show Statistics
```

Project Implementation (Hotel Reservation System)



The screenshot displays an IDE window for a project named "CodeAlpha_Hotel Reservation System". The "Project" view on the left shows the file structure, including a "Main" class. The "Main.java" file is open in the editor, showing the following code:

```
3 public class Main {
5     public static void main(String[] args) {
21         System.out.println("4. Cancel Booking");
22         System.out.println("5. Exit");
23         System.out.print("Enter Choice : ");
24
25         choice = sc.nextInt();
26         sc.nextLine();
27
28         switch (choice) {
29
30             case 1:
31                 hotel.showRooms();
32                 break;
33
34             case 2:
35                 hotel.bookRoom();
```

The "Run" view at the bottom shows the output of the program:

```
-----
HOTEL RESERVATION SYSTEM
=====
1. View Rooms
2. Book Room
3. View Bookings
4. Cancel Booking
5. Exit
Enter Choice : 1
=====
AVAILABLE HOTEL ROOMS
```

The status bar at the bottom indicates the current file is "main" in the "Main" class of the "CodeAlpha_Hotel Reservation System" project, with a timestamp of 44:27 and encoding of CRLF, UTF-8, and 4 spaces.

Learning Outcomes

Skills reinforced through practical work

- Improved confidence in writing Java programs.
- Better understanding of object-oriented programming and structured program design.
- Improved logical thinking and problem-solving ability.
- Greater familiarity with debugging and testing.
- Improved ability to understand and implement technical requirements.
- Better time management while completing assigned programming tasks.
- Experience with a practical, assignment-based development workflow.
- Improved readiness for future academic and software-development projects.

Internship Completion Certificate

Insert the official certificate here before final submission



CODE
— ALPHA —
YOUR DREAM OUR PASSION



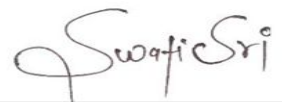
**MINISTRY OF
CORPORATE
AFFAIRS**
GOVERNMENT OF INDIA

CERTIFICATE OF COMPLETION

This Certificate Is Proudly Presented To

Aman Kumar

This Certificate recognizes the successful completion of the CodeAlpha Virtual Internship Program in
Java Programming, demonstrating dedication and sincerity throughout the program.
We appreciate the valuable efforts and wish continued success in all future endeavors.
20th July 2026 to 20th August 2026



CEO & Co-Founder

20th August 2026

Date of Issue



SCAN TO VERIFY THIS CERTIFICATE
Student ID: CA/DF1/205943

Conclusion

Overall internship outcome

Overall Outcome

The internship provided useful practical exposure to Java programming and reinforced programming knowledge through hands-on work. It encouraged independent learning, systematic debugging and disciplined task completion.

The experience strengthened the foundation for future academic and software-development projects.

References

Sources listed in the internship report

- CodeAlpha — Official Website and Internship Program Information.
- Oracle — Java Documentation and Java Platform Documentation.
- Oracle — The Java Tutorials / Learning Resources.